Convex Reductions and Averaging Obstructions for Complemented Copies in $C(\beta\omega \times \beta\omega)$

Upgraded partial packet for arXiv:2405.19120

June 22, 2026

Status

Strong partial result, likely valid, pending human review. This packet upgrades the earlier extension-operator positive subcase for source Problem 4.1. It still does not solve the problem and does not give a counterexample. It records exact reductions of the target class, positive closure principles, a concrete obstruction to the Cantor-cube averaging route, and a no-go lemma for the most direct operator-space projection.

The supplied upgrade report is preserved in the packet's `evidence/` directory. The previous active packet is preserved in the packet's `history/` directory.

1 Source Problem

The source paper is Plebanek–Rondoš–Sobota, *Complemented subspaces of Banach spaces* $C(K \times L)$, arXiv:2405.19120 [4]. Problem 4.1 asks:

Problem 4.1 (Plebanek–Rondoš–Sobota)**.** Does $C(\beta\omega \times \beta\omega)$ contain a complemented isomorphic copy of $C(K)$ for every separable compact space K?

> We start the final section of the paper by asking the following question, related to Corollary 1.4. Recall here that, for any separable compact space K, the space $\beta\omega \times \beta\omega$ maps continuously onto K and hence $\mathcal{C}(K)$ embeds isometrically into $\mathcal{C}(\beta\omega\times\beta\omega)$. We are however not aware of any example of a separable compact space K such that $\mathcal{C}(\beta\omega\times\beta\omega)$ does not contain a complemented copy of $\mathcal{C}(K)$.
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> **Problem 4.1.** *Does $\mathcal{C}(\beta\omega \times \beta\omega)$ contain a complemented isomorphic copy of $\mathcal{C}(K)$ for every separable compact space K?*
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> A natural and more general variant of Problem 4.1 could ask for a characterization of those separable compact spaces K for which $\mathcal{C}(K)$ is not isomorphic to a complemented subspace of $\mathcal{C}(\beta\omega \times \beta\omega)$.

The source paper already proves the metric compact case. More precisely, Corollary 1.4 of [4] gives complemented copies of $C([0,1]^\kappa)$, and also of $C(2^\kappa)$, inside

$$Z := C(\beta\omega \times \beta\omega) \qquad (1 \le \kappa \le \mathfrak{c}).$$

2 Definitions and Proof Intuition

For Banach spaces X, Y, write $X \overset{c}{\hookrightarrow} Y$ if there are bounded operators

$$U : X \to Y, \qquad V : Y \to X, \qquad VU = I_X.$$

Equivalently, $U[X]$ is a complemented subspace of Y isomorphic to X. The factorization constant is

$$\gamma_Y(X) = \inf\{\|U\|\|V\| : U : X \to Y,\ V : Y \to X,\ VU = I_X\}.$$

The main mechanism is transitivity: if a class of compact spaces L has $C(L) \overset{c}{\hookrightarrow} Z$, then every compact K whose $C(K)$ is complemented in some $C(L)$ is also a positive instance. The previous packet used this only for extension operators into cubes. The upgrade adds three sharper uses of the same principle.

First, every $C(K)$ is norm-one complemented in the affine functions on the probability simplex $\mathcal{P}(K)$, and also in $C(B_{\mathcal{M}(K)})$. Thus the full problem is exactly as hard on separable Bauer simplices and separable weak-star compact dual balls. Second, an averaging operator from a Cantor cube gives a complemented pullback copy, so all separable almost-Milyutin compacta are positive instances. Third, Blasco and Ivorra give a separable dyadic compactum that is not almost Milyutin; hence the positive averaging route cannot be the whole story. Finally, the operator model $Z \cong \mathcal{K}(\ell_1, \ell_\infty)$ shows that the naive projection onto $\mathcal{K}(\ell_1, C(K))$ would already imply that $C(K)$ is complemented in ℓ_∞, so any full positive proof must be more twisted.

3 Closure Principles

Lemma 3.1 (Transitivity)**.** *If $X \overset{c}{\hookrightarrow} Y$ and $Y \overset{c}{\hookrightarrow} W$, then $X \overset{c}{\hookrightarrow} W$. Moreover*

$$\gamma_W(X) \le \gamma_Y(X)\gamma_W(Y).$$

Proof. Choose $U_1 : X \to Y$, $V_1 : Y \to X$, $U_2 : Y \to W$, and $V_2 : W \to Y$ with $V_iU_i = I$. Then $(V_1V_2)(U_2U_1) = I_X$, and the stated norm estimate follows. □

Proposition 3.2 (Averaging closure)**.** *Let $q : L \to K$ be a continuous surjection between compact spaces. Suppose there is a bounded operator $A : C(L) \to C(K)$ such that*

$$Aq^* = I_{C(K)}, \qquad q^*f = f \circ q.$$

If $C(L) \overset{c}{\hookrightarrow} Z$, then $C(K) \overset{c}{\hookrightarrow} Z$.

Proof. The identity $Aq^* = I$ says that $q^*C(K)$ is complemented in $C(L)$, with projection q^*A. Apply Lemma 3.1. □

Corollary 3.3 (Previous packet as a special case)**.** *If K is compact and $C(K)$ is complemented in $C([0, 1]^\kappa)$ or in $C(2^\kappa)$ for some $1 \le \kappa \le \mathfrak{c}$, then $C(K) \overset{c}{\hookrightarrow} Z$.*

Proof. Combine the source Corollary 1.4 with Lemma 3.1. The extension-operator and retract cases in the previous packet are concrete ways to obtain the first complemented embedding. □

4 Exact Convex Reductions

For a compact space K, let $\mathcal{P}(K)$ be the weak-star compact space of probability measures on K, and let $\mathcal{M}(K) = C(K)^*$.

Theorem 4.1 (Canonical convexification)**.** *For every compact space K, the Banach space $C(K)$ is isometric to a norm-one complemented subspace of $C(\mathcal{P}(K))$.*

Proof. Define

$$\mathcal{A}_K : C(K) \to C(\mathcal{P}(K)), \qquad (\mathcal{A}_K f)(\mu) = \int_K f \, d\mu,$$

and

$$\mathcal{R}_K : C(\mathcal{P}(K)) \to C(K), \qquad (\mathcal{R}_K F)(x) = F(\delta_x).$$

The first map is an isometry because Dirac measures belong to $\mathcal{P}(K)$; the second is a contraction; and

$$(\mathcal{R}_K \mathcal{A}_K f)(x) = \int_K f \, d\delta_x = f(x).$$

Thus $\mathcal{A}_K \mathcal{R}_K$ is a norm-one projection onto $\mathcal{A}_K[C(K)]$. □

Lemma 4.2. *If K is separable compact, then $\mathcal{P}(K)$ and $B_{\mathcal{M}(K)}$ are separable compact spaces of weight at most $\mathfrak{c}$.*

Proof. Let $D \subseteq K$ be countable and dense. Finite rational convex combinations of point masses δ_d, $d \in D$, are weak-star dense in $\mathcal{P}(K)$: a basic weak-star neighbourhood only sees finitely many continuous functions, and their integrals can be approximated by rational convex combinations of values on D. For $B_{\mathcal{M}(K)}$, use finite rational convex combinations of $\pm\delta_d$. Since every member of $C(K)$ is determined by its restriction to D, one has $|C(K)| \leq \mathfrak{c}$, so the weak-star topologies involved have bases of cardinality at most $\mathfrak{c}$. □

Theorem 4.3 (Exact reduction)**.** *The following assertions are equivalent.*

(i) $C(K) \overset{c}{\hookrightarrow} Z$ *for every separable compact space K.*

(ii) $C(S) \overset{c}{\hookrightarrow} Z$ *for every separable compact convex space S.*

(iii) $C(\mathcal{P}(K)) \overset{c}{\hookrightarrow} Z$ *for every separable compact space K.*

(iv) $C(B_{\mathcal{M}(K)}) \overset{c}{\hookrightarrow} Z$ *for every separable compact space K.*

Thus any counterexample may be sought among separable Bauer simplices, or among separable symmetric weak-star compact dual balls.

Proof. The implications from (i) to (ii)–(iv) are immediate from Lemma 4.2. For (iii) implies (i), use Theorem 4.1 and transitivity. For (iv) implies (i), define

$$J_K : C(K) \to C(B_{\mathcal{M}(K)}), \qquad (J_K f)(\mu) = \mu(f),$$

and

$$S_K : C(B_{\mathcal{M}(K)}) \to C(K), \qquad (S_K F)(x) = F(\delta_x).$$

Then J_K is an isometry, $\|S_K\| \leq 1$, and $S_K J_K = I$. Hence $C(K)$ is norm-one complemented in $C(B_{\mathcal{M}(K)})$, and transitivity again gives (i). □

5 Almost-Milyutin Positive Class

Definition 5.1. A compact space K is *almost Milyutin* if there are a Cantor cube 2^Γ, a continuous surjection $q : 2^\Gamma \to K$, and a bounded averaging operator $A : C(2^\Gamma) \to C(K)$ satisfying $Aq^* = I$.

Lemma 5.2 (Coordinate reduction)**.** *Suppose $q : 2^\Gamma \to K$ is a continuous surjection onto a compact space of weight κ, and q admits a bounded averaging operator. Then there is $\Lambda \subseteq \Gamma$, $|\Lambda| \le \kappa$, such that a surjection $q_\Lambda : 2^\Lambda \to K$ also admits an averaging operator of no larger norm.*

Proof. Embed K into $[0,1]^\kappa$. Each coordinate of the composite $2^\Gamma \to K \to [0,1]^\kappa$ depends on countably many coordinates of 2^Γ. The union of those coordinate sets has cardinality at most κ; call it Λ. Then $q = q_\Lambda \pi_\Lambda$. If A averages q, define $A_\Lambda g = A(g \circ \pi_\Lambda)$. This has no larger norm and satisfies $A_\Lambda q_\Lambda^* = I$. □

Theorem 5.3. *If K is a separable almost-Milyutin compact space, then $C(K) \overset{c}{\hookrightarrow} Z$. In particular, Problem 4.1 has a positive answer for every separable Dugundji compact space.*

Proof. A separable compact space has weight at most $\mathfrak{c}$. By Lemma 5.2, the almost-Milyutin witness can be chosen as a surjection from 2^Λ with $|\Lambda| \le \mathfrak{c}$. Hence

$$C(K) \overset{c}{\hookrightarrow} C(2^\Lambda) \overset{c}{\hookrightarrow} Z,$$

where the first complemented embedding comes from the averaging operator and the second from the source Corollary 1.4. Haydon proved that every Dugundji compactum is Milyutin [2], so the final assertion follows. □

6 A Separable Obstruction to the Cantor-Cube Route

Blasco and Ivorra [1] constructed a dyadic compact space that is not almost Milyutin. Let T be the quotient of 2^{ω_1} obtained by identifying two distinct points, and put $K_* = T^{\mathbb{N}}$. For a compact space L, define

$$p(L) = \inf\{\|A\| : q : 2^\Gamma \twoheadrightarrow L,\ A : C(2^\Gamma) \to C(L),\ Aq^* = I\},$$

with $p(L) = +\infty$ if no such pair exists.

Theorem 6.1 (Blasco–Ivorra)**.** *For every positive integer n, $p(T^n) \ge n + 1$, although each finite power T^n is almost Milyutin. Moreover $K_* = T^{\mathbb{N}}$ is dyadic but not almost Milyutin; equivalently, $p(K_*) = +\infty$.*

Proposition 6.2. *The space K_* is separable, dyadic, and has weight at most $\mathfrak{c}$.*

Proof. The Hewitt–Marczewski–Pondiczery theorem implies that 2^{ω_1} is separable, since $\omega_1 \le \mathfrak{c}$; see Ross–Stone [5]. The quotient T is separable, and a countable product of separable spaces is separable. The quotient map $2^{\omega_1} \to T$ induces an onto map

$$(2^{\omega_1})^{\mathbb{N}} \cong 2^{\omega_1 \times \mathbb{N}} \cong 2^{\omega_1} \longrightarrow T^{\mathbb{N}},$$

so K_* is dyadic. The weight estimate is immediate. □

Proposition 6.3 (Uniform finite-stage test)**.** *If $C(K_*) \overset{c}{\hookrightarrow} Z$, then*

$$\sup_{n \ge 1} \gamma_Z(C(T^n)) < \infty.$$

In fact $\gamma_Z(C(T^n)) \le \gamma_Z(C(K_*))$ for every n.

Proof. Fix $t_0 \in T$. The coordinate projection $K_* \to T^n$ gives an isometric embedding $J_n : C(T^n) \to C(K_*)$. Evaluation at the tail $(t_0, t_0, \dots)$ gives a contraction $R_n : C(K_*) \to C(T^n)$ with $R_n J_n = I$. Composing any factorization of $I_{C(K_*)}$ through Z with J_n and R_n gives the asserted bound. □

Thus a negative answer would follow from a comparison

$$p(T^n) \le \Phi(\gamma_Z(C(T^n)))$$

for a finite-valued function Φ independent of n. Such a comparison is not proved here. This is precisely where the obstruction stops short of a counterexample: $p(T^n)$ measures canonical averaging through Cantor cubes, while γ_Z allows arbitrary isomorphic complemented copies.

7 Operator-Space Reformulation and No-Go Lemma

There are natural isometric identifications

$$Z \cong C(\beta\mathbb{N}, C(\beta\mathbb{N})) \cong C(\beta\mathbb{N}, \ell_\infty) \cong \mathcal{K}(\ell_1, \ell_\infty) \cong \ell_\infty \widehat{\otimes}_\varepsilon \ell_\infty.$$

For the middle identification, an operator $S : \ell_1 \to \ell_\infty$ is represented by the sequence (Se_n); compactness is exactly relative norm compactness of this range sequence.

If K is separable and (d_n) is dense in K, then

$$E_K : C(K) \to \ell_\infty, \qquad E_K f = (f(d_n))_{n \in \mathbb{N}},$$

is an isometry. The direct strategy would be to place $\mathcal{K}(\ell_1, E_K C(K))$ inside $\mathcal{K}(\ell_1, \ell_\infty)$ and project onto it. The next lemma shows that this strategy cannot give new cases.

Lemma 7.1 (Operator-space no-go)**.** *Let X be a closed subspace of a Banach space Y. If $\mathcal{K}(\ell_1, X)$ is complemented in $\mathcal{K}(\ell_1, Y)$, then X is complemented in Y. Moreover, a projection of norm M in the operator space produces a projection of norm at most M in Y.*

Proof. For $y \in Y$, let $R_y : \ell_1 \to Y$ be the rank-one operator $R_y(a) = a_1 y$. If $Q : \mathcal{K}(\ell_1, Y) \to \mathcal{K}(\ell_1, X)$ is a projection, set

$$P(y) = Q(R_y)e_1.$$

Then $P : Y \to X$ is bounded with $\|P\| \le \|Q\|$. If $x \in X$, then $R_x \in \mathcal{K}(\ell_1, X)$, so $Q(R_x) = R_x$ and $P(x) = x$. Hence P is a projection onto X. □

8 Fibre-Supported Coupling Criterion

Proposition 8.1. *Let $\rho : L \to K$ be a continuous surjection between compact spaces. Suppose there is a weak-star continuous map*

$$K \to \mathcal{P}(L), \qquad x \mapsto \nu_x,$$

such that $\operatorname{supp} \nu_x \subseteq \rho^{-1}(x)$ *for every $x \in K$. Then*

$$Af(x) = \int_L f \, d\nu_x$$

defines a norm-one positive averaging operator for ρ. Consequently, if $C(L) \overset{c}{\hookrightarrow} Z$, then $C(K) \overset{c}{\hookrightarrow} Z$.

Proof. Weak-star continuity gives $Af \in C(K)$, and $\|A\| \le 1$. If $g \in C(K)$, then

$$A(g \circ \rho)(x) = \int_{\rho^{-1}(x)} g(\rho(y))\, d\nu_x(y) = g(x).$$

Apply Proposition 3.2. □

The source paper's compact-group construction supplies this criterion in a highly structured situation. A general replacement for the group operation is not known from the present packet.

9 Verification Notes and Limitations

The mathematical classification is partial. The exact reductions and closure principles are operator-theoretic and topological consequences of norm-one complemented embeddings. The positive class is genuinely larger than the previous extension-operator packet because it records the full almost-Milyutin/Dugundji mechanism, but it still does not cover all separable compact spaces.

The Blasco–Ivorra space K_* is not a counterexample to Problem 4.1; it is a counterexample only to the universal Cantor-cube averaging route. The operator-space no-go lemma blocks only the direct projection onto $\mathcal{K}(\ell_1, E_K C(K))$, not arbitrary twisted copies of $C(K)$ in Z.

Local duplicate checks on June 22, 2026 found the earlier partial packet and the full-general no-promotion attempt, but no packet already recording the convex reductions, the Blasco–Ivorra finite-stage test, or the no-go lemma. The supplied report records an external literature search through June 20, 2026 and states that no full answer or counterexample was located; this upgrade did not perform a fresh external web search.

10 Human Review Recommendation

Reviewers should check:

1. the separability and weight estimates in Lemma 4.2;
2. the use of Haydon's Milyutin/Dugundji theorem in Theorem 5.3;
3. the precise Blasco–Ivorra statement quoted in Theorem 6.1;
4. that Lemma 7.1 is used only as a route obstruction.

References

[1] J. L. Blasco and C. Ivorra, *On dyadic spaces and almost Milyutin spaces*, Colloq. Math. **67** (1994), 241–244.

[2] R. Haydon, *On a problem of Pełczyński: Milutin spaces, Dugundji spaces and AE(0-dim)*, Studia Math. **52** (1974), 23–31.

[3] A. Pełczyński, *Linear extensions, linear averagings, and their applications to linear topological classification of spaces of continuous functions*, Dissertationes Math. **58** (1968), 1–89.

[4] G. Plebanek, J. Rondoš, and D. Sobota, *Complemented subspaces of Banach spaces $C(K \times L)$*, J. Funct. Anal. **290** (2026), no. 2, Paper No. 111236; arXiv:2405.19120.

[5] K. A. Ross and A. H. Stone, *Products of separable spaces*, Amer. Math. Monthly **71** (1964), 398–403.